

DEERE & COMPANY (JOHN DEERE) AT THE CROSSROADS

Betting the Farm on Software

A Comprehensive Digital Transformation Case Study

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MAIN CASE NARRATIVE

I. OPENING: A MOMENT OF DECISION

"It's All About Doing More with Less"

On the morning of November 26, 2025, John C. May, Chairman and Chief Executive Officer of Deere & Company, delivered news that would have been almost unthinkable two years earlier: the world's largest agricultural equipment manufacturer was guiding to fiscal 2026 net income of just \$4.0 to \$4.75 billion, well below analysts' \$5.33 billion consensus and roughly half of the \$10.2 billion the company had earned in fiscal 2023. Quarterly profit had fallen to \$1.06 billion from \$1.24 billion a year earlier. The company faced approximately \$600 million in direct tariff impact in 2025, with continued pressure expected in 2026.

And yet, in the same breath, May made a remarkable statement: 2026, he said, "could mark the bottom of the agricultural cycle." He was not signaling retreat. He was signaling that, even as Deere's traditional equipment business faced a sharp downturn, the company would accelerate its decade-long bet on a digital transformation, a bet to reinvent a 187-year-old hardware company into something closer to Apple, with software, subscriptions, and recurring revenue at its core.

The tension is the case. May was simultaneously asking shareholders to absorb a 30 percent profit decline, asking farmers to absorb price increases on machines they could increasingly not afford, and asking both groups to trust that the same proprietary software which had triggered a \$99 million class action settlement and an active Federal Trade Commission (FTC) lawsuit would also be the foundation of Deere's next \$150 billion of customer value. The contradiction was not lost on him.

"It's all about doing more with less," May had said three years earlier when he first publicly committed Deere to a software-led future. By the end of the decade, he projected that 10 percent of Deere's annual revenue would come from software fees. In 2025, the company conceded in its 10-K that "revenue from SaaS products did not represent a significant percentage of our revenues."

How May and his team arrived at this moment, and what they would do next, is the question this case asks.

II. THE PROTAGONIST: AN INSIDER REINVENTING AN ICON

John C. May is a longtime Deere insider rather than a Silicon Valley import. He served as President and Chief Operating Officer in 2019, became Chief Executive Officer and President later that same year, and assumed the role of Chairman in 2020. At his earliest CES appearance as CEO, he reminded the audience that Deere had been investing in the foundational layers of precision agriculture — guidance, telematics, and data management, for "the last 20 years" and had begun working with progressive farmers on its precision ag journey "in 2012."

Within twelve months of becoming CEO, May reorganized the company. The "Smart Industrial Operating Model," unveiled in 2020, replaced Deere's old product-centric structure (a tractor business, a sprayer business, a harvester business) with three pillars built around the customer's job to be done: Production Systems, the Technology Stack, and Lifecycle Solutions.

Smart Industrial Operating Model — Three Pillars:

Pillar	Definition	Strategic Goal
Production Systems	Organized by what farmers actually grow (corn, small grains, cotton, sugarcane)	Deep customer-centric focus on growers' P&L
Technology Stack	Embedded software, sensors, connectivity, and cloud platform across all machines	Recurring revenue via SaaS and data
Lifecycle Solutions	Integrated parts, service, and software upgrade business	Monetize the full ownership experience, not just the original sale

In 2022, May layered the "Leap Ambitions" on top, a set of long-term financial and operational goals oriented around the use of "differentiated equipment and service solutions, including automation, autonomy, digitalization, lifecycle solutions, and Solutions as a Service (SaaS)." In December 2025, with the cycle deteriorating, May refined those Ambitions rather than abandoning them.

Justin Rose, the former Boston Consulting Group senior partner who joined Deere in October 2022 as President of Lifecycle Solutions, articulated the strategic intuition on a Technology and Services Industry Association (TSIA) podcast: the realization, "about 10 years ago," was that Deere "needed to move away from a very equipment-centric worldview... to a more customer-centric point of view where we say, you know, a customer doesn't tractor, the customer grows corn." In Rose's later formulation to OpenAI, Deere was "weaving AI into every stage", building "an AI-native customer success model that could be applied to any industry."

III. THE TWO BETS: SEE & SPRAY AND AUTONOMY

Bet One: See & Spray - Optimizing the Spray

In 2017, Deere acquired Blue River Technology, a startup co-founded in 2011 by Peruvian immigrant Jorge Heraud, for \$305 million. Heraud's pitch was simple and arresting: about two-thirds of the herbicide applied by a conventional broadcast sprayer lands on bare soil or healthy crops rather than weeds. According to Bloomberg NEF research, as little as 1 percent of pesticides applied via broadcast actually reach the target pest. American farmers spend roughly \$25 billion annually on 3 billion pounds of herbicide, with a per-acre application cost of \$50 to \$100.

Deere's See & Spray system, built on Blue River's core technology, attacks this waste directly. Boom-mounted cameras scan 2,100 square feet of crop area per second, feeding images into a deep neural network running on NVIDIA Jetson Xavier processors capable of "tens of trillions of operations per second", classifying each pixel as crop or weed and triggering the nozzles only over weeds. On a 120-foot spray boom, 36 cameras feed onboard computers in real time.

See & Spray Performance Results:

Metric	Result	Source
Herbicide saved (2024)	8 million gallons	Deere 2024 News Release
Average herbicide savings	59%	Deere 2024 News Release
Iowa State field study average	76% reduction (43.9%–90.6%)	Iowa State University (415 acres)
Direct chemical cost savings	\$15.7 per acre / \$6,500 per field	Iowa State University
Deere ExactApply Nebraska/Kansas trials (2025)	Up to 77% herbicide savings	Mordor Intelligence 2026

Just as important as the technology is the business model. See & Spray "pioneered a shift in John Deere's pricing model": the company moved to subscription-based, renewable licenses, "invoicing customers solely based on the acres where the technology is actively used. Farmers are only billed after spraying operations, aligning costs directly with the tangible value received." For the farmer, this lowers upfront risk; for Deere, it creates the recurring revenue stream that May has promised investors.

Bet Two: Autonomy- Replacing the Driver

In August 2021, Deere paid \$250 million for Bear Flag Robotics, a Silicon Valley startup specializing in retrofitting conventional tractors with sensors and software for full autonomy. It was Deere's second major Silicon Valley acquisition in four years and a clear signal that autonomous tractors would be central to the future of farming.

At CES 2022, Deere unveiled the 8R autonomous tractor. At CES 2025, the company expanded the lineup to two additional autonomous tractors, an autonomous quarry dump truck, and a battery-electric commercial mower, plus a second-generation autonomy kit that retrofits existing equipment. A Tech Buzz reviewer called Deere's CES 2026 booth "one of the strongest real-world technology showings on the floor."

The technology is dense. The autonomous platform combines 6–16 cameras, sub-inch GPS accuracy enabled by a proprietary GNSS network Deere began building in 1998-1999 ("we're one of only five companies that actually have the capability to do the geolocation and precision at this level," Rose said), and a deep neural network trained on hundreds of thousands of farm images that classifies obstacles in roughly 100 nanoseconds on dual onboard GPUs.

Autonomous Tractor Business Impacts:

Impact Area	Measured Outcome	Source
Farm productivity	15–20% productivity gain after adoption	GriNext Conference / AI at JD
Fuel & labor efficiency	6% average reduction in fuel and labor costs	AI at John Deere (2025)
Labor gap addressed	Replaces 12+ hour skilled operator shifts	CTO Jahmy Hindman, Reuters 2025

Impact Area	Measured Outcome	Source
Farm labor shortage context	~2.4 million farm jobs unfilled annually in the U.S.	American Farm Bureau Federation

Deere executives increasingly framed the rationale for autonomy in demographic terms, citing labor constraints alongside technological capability. Jahmy Hindman stated at CES 2025: “Our customers have work that must get done at certain times of the day and year, yet there is not enough available and skilled labor to do the work.” The average American farmer is over 58 years old and often works 12-to-18-hour days. The United States Department of Agriculture reported that the number of U.S. farms declined by 7 percent between 2017 and 2023. In parts of the U.S. Midwest, a planting window of approximately ten days can determine whether a farm completes its seasonal operations.

IV. THE APPLE PLAYBOOK AND ITS DISCONTENTS

In May 2022, Reuters reported that Deere was explicitly ‘borrowing a page from big technology firms like Apple,’ aiming to boost revenue by combining its traditional machinery business with software and subscription-based services. A 2021 Bernstein analyst report estimated average gross margins for farming software at approximately 85 percent, compared to 25 percent for equipment sales. By 2026, Deere stated that it intended to connect 1.5 million machines and 500 million acres to the John Deere Operations Center cloud platform. Justin Rose described the potential opportunity as approximately \$150 billion of value to be unlocked through technology applied to Deere’s existing installed base.

“The Apple-like approach also introduced challenges similar to those faced by technology firms, particularly around control, data ownership, and repair access. Farmers had historically modified and repaired their own equipment independently. As tractors became increasingly software-driven, often described as ‘computers on four wheels’, Deere’s proprietary systems limited certain repairs to authorized dealers. Walter Schweitzer told the Wall Street Journal in 2022: ‘You’re losing control of the data, and the ownership of the tools.’”

Right-to-Repair: Timeline of Key Legal and Regulatory Events

Date	Event
June 2022	13 lawsuits consolidated in federal court in Illinois over repair monopolization
January 2023	Deere signs MOU with American Farm Bureau Federation on repair access
January 2025	FTC and 5 U.S. state AGs file antitrust lawsuit against Deere over repair policies
April 2026	Deere settles class action for \$99 million; commits to 10-year repair tool access

“Repair advocates estimated that Deere’s restrictions may have resulted in total costs to farmers of up to \$4.2 billion, with antitrust economist Russell Lamb estimating overcharging alone at between \$190 million and \$387 million. Nathan Proctor of the US PIRG told Wired in 2026 that advocacy groups remained skeptical, citing prior industry patterns.”

Industry analyst Sangeet Paul Choudary of UC Berkeley described this shift in his 2026 book *Reshuffle*: “By moving decision-making away from the farmer and into the machine, Deere alters the logic of value creation and, consequently, the organizational system of the farm. In doing so, it shifts power from the farmer to the tool.”

For May, the open question is whether the trust required to sell software at 85 percent gross margin can survive a multi-year legal narrative in which Deere is cast as the company taking power away from farmers.

V. THE CYCLE BITES: FINANCIAL PRESSURE MEETS DIGITAL AMBITION

John Deere Key Financial Metrics (2023–2026):

Metric	FY2023 (Peak)	FY2024	FY2025	FY2026 Guidance
Total Net Sales & Revenues	\$61.3 B	\$51.7 B	\$45.7 B	—
Net Income	\$10.2 B	\$7.1 B	\$5.0 B	\$4.5–\$5.0 B (revised up)
P&PA Operating Margin	~26%	21.7%	15.4%	Under pressure
R&D Expenditure	~\$2.3 B	\$2.29 B	\$2.31 B	Maintained
Tariff Impact (pre-tax)	—	—	~\$600 M	~\$1.2 B expected
Total Employees	~82,000	~75,800	~73,100	—

By November 2025, the strategic tensions had collided with a brutal macro environment. Production & Precision Agriculture sales fell 17 percent, with operating margin compressing from 21.7 percent to 15.4 percent. U.S. and Canada PPA volumes fell 30 percent. Construction & Forestry sales dropped 12 percent on weak commercial construction and elevated dealer used-equipment inventories.

Yet on February 19, 2026, Deere raised full-year FY2026 net income guidance from \$4.0–\$4.75 billion to \$4.5–\$5.0 billion, after Q1 revenue of \$9.61 billion (up 13 percent year-over-year) beat expectations. Construction equipment and small agriculture equipment were showing signs of recovery. Deere shares rose 8 percent on the announcement. The cycle may be bending, but the strategic question remains open.

VI. STRATEGIC TENSIONS: THREE UNRESOLVED QUESTIONS

1. Subscription Growth vs. Farmer Trust

The Apple-like recurring revenue model is the financial engine of May's transformation. But each year of right-to-repair litigation, each FTC filing, each Wired headline, makes the underlying customer relationship more adversarial. Can Deere build the trust required for farmers to welcome a tractor that "shifts power from the farmer to the tool"? Deere's 2025 Annual Report openly lists adoption and monetization of technology as strategic risks, "customers may not understand the value proposition."

The settlement commits Deere to 10 years of repair tool access. After that, Deere could simply choose to change how it manages repair access.

2. Investment vs. Discipline in a Downturn

In FY2025, Deere spent \$2.31 billion on R&D, essentially flat year-over-year and slightly higher than its peak-revenue-year R&D, and \$1.30 billion on capital expenditures, while year-end headcount fell by approximately 2,700 employees following \$157 million in pre-tax employee separation programs. With used equipment inventory still elevated and tariff costs of \$1.2 billion expected in 2026, every dollar that goes into autonomous tractor R&D is a dollar that does not buy back stock or pay dividends. Wall Street has historically rewarded Deere for capital returns; it may not yet reward Deere for software platforms that still don't drive material revenue.

3. The Dealer Dilemma

Deere's dealer network, roughly 2,050 independent locations in the U.S. and Canada alone is, in Rose's words, "one of our most significant competitive advantages." But the SaaS, autonomy, and direct-data-relationship strategies all run through that dealer channel and risk disintermediating it. Rose has launched a new global customer success function and a dealer business system rollout with "Expert Connect" software that uses ChatGPT to summarize inbound service calls. But the historical lesson from enterprise software is unavoidable: "some of the channel partners will make it and some won't."

It is early 2026. As John C. May reviews Deere's performance and strategic trajectory, he faces a set of difficult choices. Should the company continue to aggressively invest in autonomy and SaaS despite declining profits? Should it reconsider its approach to software control in response to regulatory pressure and farmer backlash? Or should it prioritize near-term financial recovery, even if it slows the digital transformation?

The decisions he makes in the coming months may determine whether Deere successfully evolves into a technology-driven platform company, or remains constrained by the limits of its traditional business model.

VII. DISCUSSION QUESTIONS

1. What are the strategic advantages and risks of John Deere transitioning from a traditional hardware manufacturer to a SaaS and data-driven technology company? How does this resemble, and differ from, the Apple model?
2. How does the FTC's antitrust lawsuit regarding right-to-repair threaten Deere's Smart Industrial Operating Model and Technology Stack? Is Deere's legal exposure existential or manageable?
3. Given the macroeconomic downturn of 2025 (dropping net sales and income from \$61.3B to \$45.7B), should John Deere accelerate or slow down its investments in AI and autonomous technologies?
4. Can Deere simultaneously maintain its dealer network's trust and loyalty while building the direct-data and SaaS relationships that its recurring-revenue model requires?

5. How can John Deere balance the necessity of protecting its intellectual property (source code, AI models) with the farmers' critical need to repair their machinery quickly during short harvesting windows?
6. If you were John C. May in early 2026, what would you do differently to protect both the near-term earnings recovery and the long-term digital transformation thesis?

TECHNICAL SUPPLEMENT

INDUSTRY ECONOMICS & MARKET TRENDS

The North America precision agriculture market was valued at \$4.10 billion in 2025 and is projected to reach \$6.05 billion by 2031, reflecting a 6.7% compound annual growth rate (CAGR). The agricultural industry is highly cyclical and depends heavily on farm income, global trade policies, and commodity prices. While hardware currently dominates the precision agriculture sector with a 68.0% market share, software is the fastest-growing segment, projected to grow at a 9.8% CAGR through 2031.

Market Driver	Impact on CAGR	Timeline
Optimized variable-rate inputs lowering costs	+1.4%	Medium term (2–4 years)
Subsidies promoting smart equipment use	+1.2%	Short term (≤ 2 years)
Increased adoption of IoT sensors & analytics	+1.3%	Medium term (2–4 years)
Carbon-credit gains from precise nitrogen use	+0.9%	Long term (≥ 4 years)
Private 5G networks on large farms	+0.6%	Long term (≥ 4 years)
High capital costs (restraint)	–1.1%	Short term (≤ 2 years)
Data interoperability & privacy issues (restraint)	–0.7%	Medium term (2–4 years)

COMPETITIVE LANDSCAPE

The global agricultural and heavy equipment market is highly competitive and concentrated. John Deere's primary competitors include CNH Industrial N.V., AGCO Corporation, Trimble Inc., and Topcon Corporation. The North America precision agriculture market is moderately concentrated in 2025, with the top five companies being Deere & Company, Trimble Inc., CNH Industrial N.V., AGCO Corporation, and Topcon Corporation.

Competitor	Digital Strategy	Competitive Threat to Deere
CNH Industrial	Pay-per-acre packages integrating telematics & agronomy support	Targets Deere's capital-constrained customers
AGCO / Fendt	FendtONE platform; pre-installed 5G modems; autonomous tractors for specialty crops (2025)	Strong in European markets, growing in North America

Competitor	Digital Strategy	Competitive Threat to Deere
Trimble Inc.	Open APIs, mixed-fleet management; joint venture with AGCO (2023)	Brand-agnostic data dashboards appeal to mixed-fleet operators
Kubota + Agtonomy	Autonomous narrow tractors for organic/specialty crops (2025)	Niche autonomy play in specialty crops
TeeJet Technologies	DynaJet nozzle at lower price vs. ExactApply; comparable weed control	Price competition on precision spraying

In May 2025, Deere acquired Sentera, Inc. to integrate aerial field scouting capabilities (drone imagery and analytics) into its Operations Center digital ecosystem, a move that strengthened its analytical edge but also signaled ongoing investment during a cyclical downturn.

CRITICAL TECHNOLOGY CONCEPTS

See & Spray Technology Architecture:

Component	Specification
Cameras	36 cameras on 120-ft boom; up to 16 on autonomous tractors; full 360° field of view
Scanning speed	2,100 sq ft of crop area per second
Processing	NVIDIA Jetson Xavier GPUs; tens of trillions of operations per second
AI system	Deep neural network (convolutional); pixel-level crop/weed classification
Connectivity	Cloud-based John Deere Operations Center; continuous learning from field data
GPS precision	Sub-inch accuracy; proprietary GNSS network (one of only 5 global providers)

KEY TECHNOLOGY ACQUISITIONS

Acquisition	Year	Price	Technology	Outcome
Blue River Technology	2017	\$305 M	Computer vision, ML for crop/weed ID	See & Spray flagship product; subscription pricing model
Bear Flag Robotics	2021	\$250 M	Autonomous tractor systems, sensors	8R autonomous tractor; CES 2025 lineup; retrofit autonomy kits
Sentera Inc.	2025	N/A	Drone imagery & field analytics	Integrated into JD Operations Center; aerial scouting

GLOSSARY

Term	Definition
Smart Industrial Operating Model	Deere's 2020 strategic realignment from product-centric to customer-centric structure around Production Systems, Technology Stack, and Lifecycle Solutions
SaaS (Software as a Service)	Software licensed on subscription basis (e.g., Deere charging per acre for See & Spray usage after spraying operations)
See & Spray	Deere's AI-powered precision spraying system that detects weeds via computer vision and sprays only targeted plants
Right-to-Repair	Consumer rights movement and legal framework demanding equipment owners and independent mechanics have access to software and tools needed to repair modern machinery
GNSS (GPS)	Global Navigation Satellite System; Deere's proprietary network enables sub-inch accuracy, one of only 5 global providers at this precision level
Leap Ambitions	Deere's long-term strategic goals (2022) oriented around automation, autonomy, digitalization, lifecycle solutions, and SaaS
Operations Center	John Deere's cloud-based farm management platform intended to connect 1.5 million machines and 500 million acres
Variable-Rate Technology (VRT)	Precision farming technology enabling growers to adjust seed, fertilizer, and pesticide application rates based on field micro-zones

ADDITIONAL SOURCES & BIBLIOGRAPHY

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KEY QUOTES BY THEME

On the Core Strategy

"It's all about doing more with less.", John C. May, CEO, Deere & Company

On Business Model Transformation

"Instead of having a tractor business and a sprayer business... we have a business that's organized around what we call production systems... we have true experts that understand the growers' P&L.", Justin Rose, President of Lifecycle Solutions

On AI Integration

"It's all about weaving AI into every stage. We're building an AI-native customer success model that could be applied to any industry.", Justin Rose (via OpenAI blog, cited in Forbes 2026)

On Labor & Autonomy

"Our customers have work that must get done at certain times of the day and year, yet there is not enough available and skilled labor to do the work.", Jahmy Hindman, Chief Technology Officer

On Power Shift

"By moving decision-making away from the farmer and into the machine, Deere alters the logic of value creation and, consequently, the organizational system of the farm. In doing so, it shifts power from the farmer to the tool.", Sangeet Paul Choudary, UC Berkeley (Reshuffle, 2026)

On Right-to-Repair

"We're not going to stop advocating for farmers until their equipment belongs to them and does what they want it to, and is not beholden to whatever John Deere deems as an interest to shareholders.", Nathan Proctor, US PIRG

On Market Cycle

"We believe 2026 could mark the bottom of the agricultural cycle.", John C. May, Q4 FY2025 earnings call, November 26, 2025

FULL EVENT TIMELINE

Year	Event
1837	John Deere begins manufacturing steel plows in Grand Detour, Illinois
1998–1999	Deere begins building proprietary GNSS satellite network (sub-inch GPS precision)
2012	Deere begins working with progressive farmers on precision ag journey (AutoTrac, telematics)
2017	Deere acquires Blue River Technology for \$305M- computer vision/ML for agriculture
2019	John C. May becomes CEO and President of Deere & Company
2020	Smart Industrial Operating Model unveiled; May assumes Chairman role
2021 Aug	Deere acquires Bear Flag Robotics for \$250M- autonomous tractor systems
2022 Jan	8R autonomous tractor unveiled at CES 2022
2022 Jun	13 right-to-repair lawsuits consolidated in federal court in Illinois
2022 Sep	WSJ reports Deere targets 10% of revenue from software fees by end of decade
2022 May	Reuters: Deere adopts Apple-like tech/services model
2023 Jan	Deere signs MOU with American Farm Bureau Federation on repair access
2023	Leap Ambitions articulated; FY2023 revenue peaks at \$61.3B; net income \$10.2B
2024 Nov	Deere guides FY2025 net income \$5.0–\$5.5B- below analyst estimate of \$5.93B
2025 Jan	FTC + 5 state AGs file antitrust lawsuit against Deere over repair monopoly
2025 Jan	CES 2025: Two new autonomous tractors, autonomous dump truck, battery-electric mower unveiled
2025 May	Deere acquires Sentera Inc. - drone imagery & analytics integrated into Operations Center
2025 Nov	Q4 FY2025: Net income \$1.06B; annual net income \$5.0B; guides FY2026 \$4.0–\$4.75B
2026 Feb	Deere raises FY2026 guidance to \$4.5–\$5.0B after Q1 revenue beats expectations (+13% YoY)
2026 Apr	Deere settles class-action right-to-repair lawsuit for \$99M; commits to 10-year repair access